

Deep Space Communication utilizing IP Stack Algorithm

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ABSTRACT

The spike in the number of satellites being launched into orbit every year due to commoditization of space has led to traditional RF frequency bands used by these satellites, such as X-band to suffer from frequency congestion and delays. Existing technologies such as IP Stack can be adapted for deep space communication. This paper will suggest better communication methods which uses new state of the art technology utilizing IP stack for enhanced Earth-Mars Earth-Moon and Moon-Mars communication. The paper will also delve into the implementation of reinforcement learning based delay tolerant networks and a demonstration of protocols such as SCPS-TP and IPv6.

OBJECTIVES

- Develop an IP-based communication model for deep space using modern protocols like IPv6, UDP, and QUIC, thus implement SCPS-TP over IPv6 to ensure reliable packet parsing across long-delay links.
- Analyze and compare link budgets with Monte Carlo simulated results.
- Apply reinforcement learning (SARSA- λ) for intelligent routing in delay-tolerant network (DTN).
- Also integrate anomaly detection to identify communication disruptions using isolation forest.
- Evaluate performance metrics across multiple routing protocols.
- Our aim is a future-proof deep-space communication framework.

METHODOLOGY

Antenna Specifications:

Ka-Band Parabolic Reflector Antenna implemented with MATLAB's Antenna Designer toolbox. (See Table I)

Antenna Type	Parabolic Reflector
Frequency	32 to 34.2 GHz
Diameter	1.5 m
Focal Length	0.7
f/d ratio	0.467 m
Exciter	Conical horn
Aperture radius of the exciter	0.102 m

Table I: Antenna Specifications

Application Layer	HTTP	Media	Tunnel	Apps	Apps
Transport Layer	QUIC (+TLS)	COAP	UDP	UDP	Apps
Network Layer					
Physical Layer	CCSDS Space Links	802.3-11/WiFi			3GPP

Fig. 1: IP Stack for Deep Space Communication

Link Case Studies (Fig. 2):

Case A: Earth-Moon direct link.
Case B: Earth-Mars direct link.
Case C: Moon-Mars direct link.

Emphasis on direct satellite-to-satellite communication.

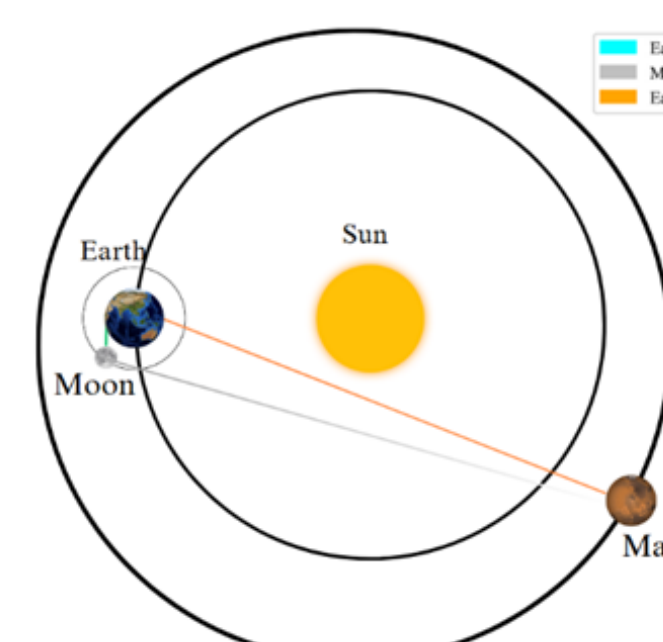


Fig. 2: Interplanetary Links

- Table II present the theoretical link budget analysis carried out based on the data from the antenna.
- The received power is well within the acceptable threshold for all the three cases.

Parameters	Earth to Moon		Earth to Mars		Moon to Mars	
	Up-Link	Down-Link	Up-Link	Down-Link	Up-Link	Down-Link
Distance (m)	3.84×10^8	3.84×10^8	4.00×10^{11}	4.00×10^{11}	4.014×10^{10}	4.014×10^{10}
Frequency (GHz)	34.2	32	34.2	32	34.2	32
FSPL (dB)	234.81	234.23	277.43	294.59	275.20	274.59
Data Rate (Mbps)	4.5	4.5	3	3	4.5	4.5
Transmitter Power (dBm)	73	50	73	50	50	60
Transmitter Gain (dBi)	47	46.7	47	46.7	47	46.7
Receiver Gain (dBi)	47	47.6	47	47.6	47	47.6
Losses (dB)	4	4	2	2.71	4	4
Receiver Power (dBm)	-79.21	-101.33	-112.43	-153.19	-135.20	-124.89

Table II: Link Budget Analysis

REALISTIC CONSTRAINTS

- Extreme propagation delays due to vast interplanetary distances particularly in the Earth-Mars link pose a major hurdle. These delays the management of acknowledgments and retransmission strategies.
- Lack of real-time space communication data, as agencies like NASA and SpaceX keep their mission data highly secured and inaccessible to the public.

ENGINEERING STANDARDS

- ITU-R SA.2167 standard
- IETF RFC 8200 IPv6
- SCPS-TP Space Communication Protocol Specification-Transport Protocol

RESULTS AND DISCUSSION

Monte Carlo Scatter Analysis:

- Monte Carlo Simulation models randomness in signal propagation. Simulated values closely follow the theoretical curve, showing high correlation.
- The scatter plots shows minimal variation from the theoretical curve (Fig.3). The histogram peak lies in the theoretical received power value (Fig. 4)

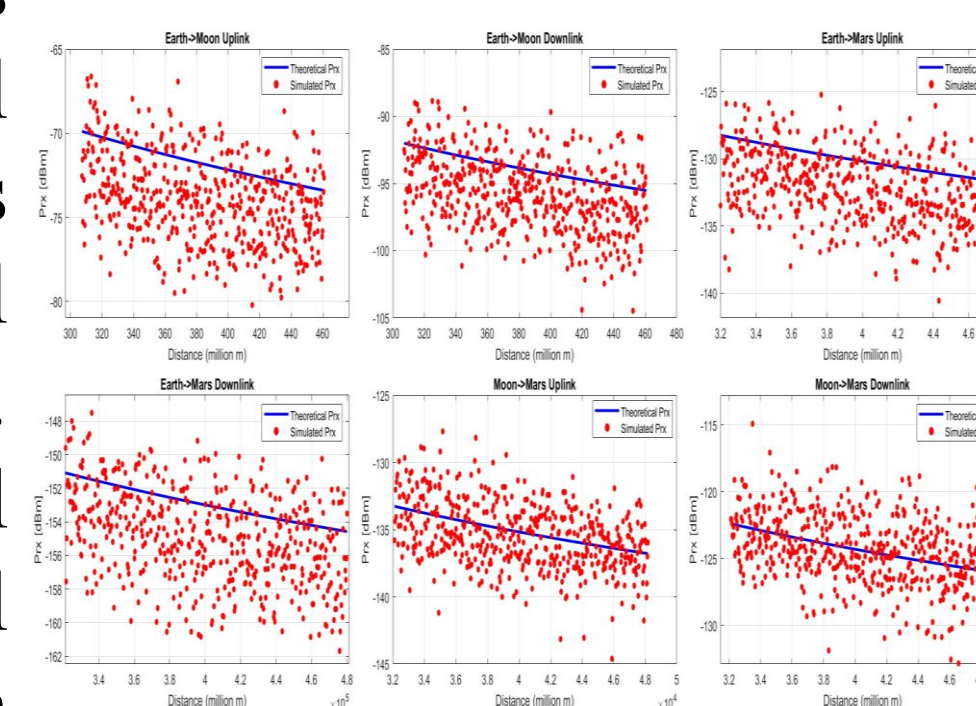


Fig. 3: Monte Carlo Scatter plot

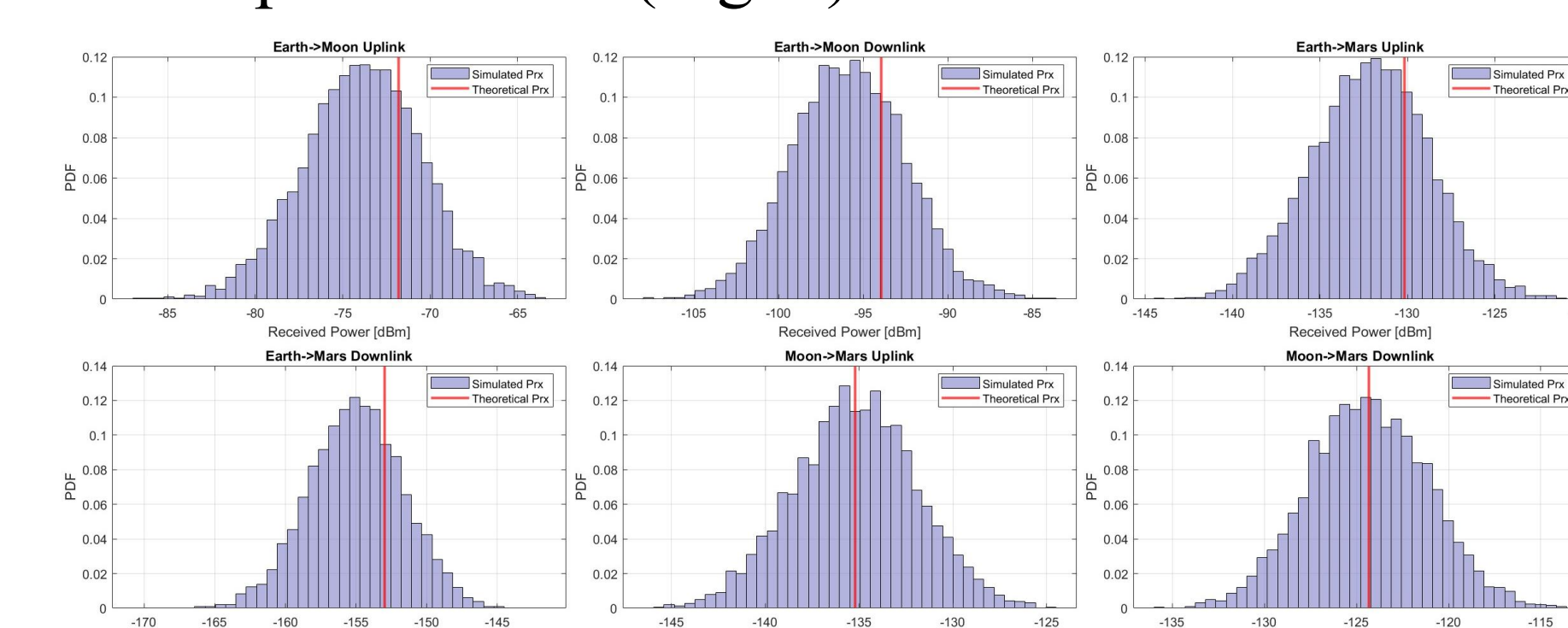


Fig. 4: Monte Carlo Histogram

SCPS/IPv6 Implementation:

- SCPS (9-byte) encapsulated in IPv6 (40-byte). Successfully builds, parses and recovers "Hello from deep space! This is a test message..."
- Larger Payloads reduce overhead, improving efficiency.

Payload (Bytes)	SCPS Packet (Bytes)	IPv6 Packet (Bytes)	SCPS overhead (%)	IPv6 overhead (%)
16	39	79	59.0	50.6
64	87	127	26.4	31.5
256	279	319	8.2	12.5
1024	1047	1087	2.2	3.7
4069	4119	4159	0.6	1.0

Table III: SCPS/IPv6 Overhead results

SCPS-TP Performance vs LTP Performance (Fig. 5 & 6):

SCPS-TP's superior performance stems from its advanced congestion control mechanisms, likely including selective negative acknowledgments (SNACKs) and rate-based control strategies.

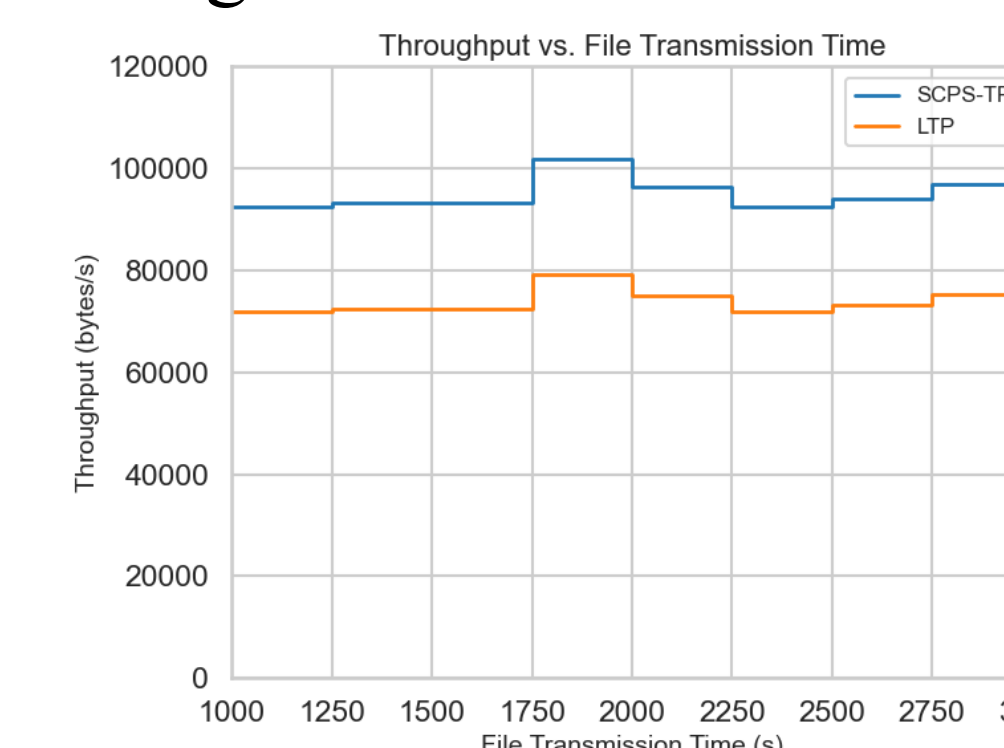


Fig. 5: Throughput vs File Transmission Time

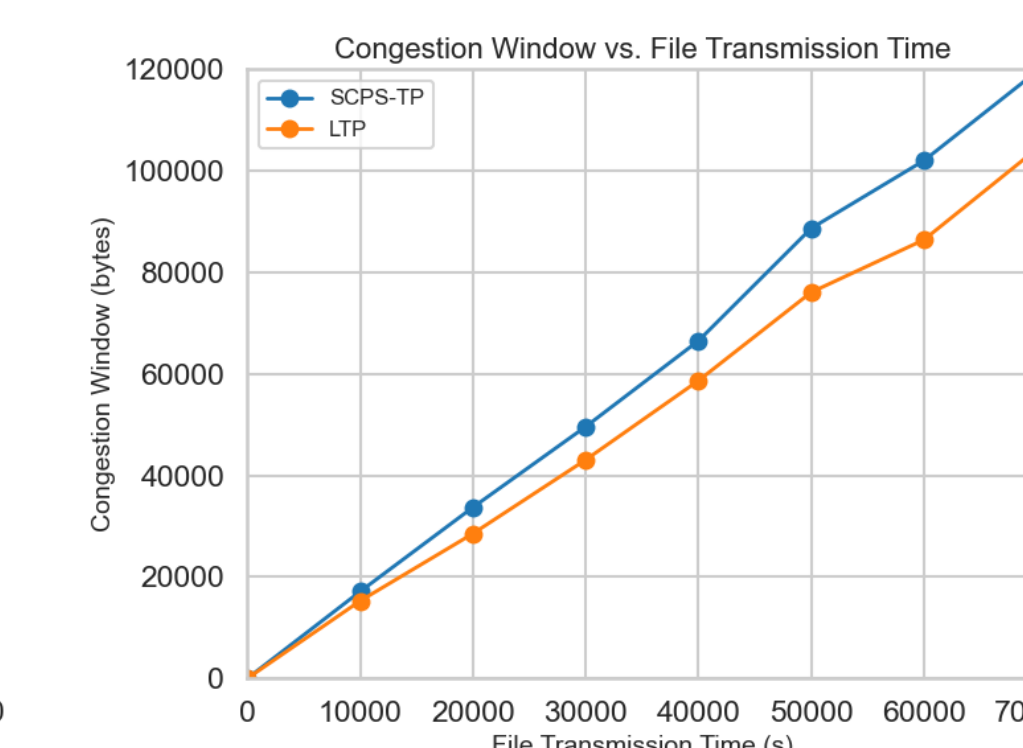


Fig. 6: Congestion Window vs File Transmission Time

RL BASED DTN

Reinforcement Learning based DTN (Fig. 7):

- Modelled in Python, optimizes routing using a reward-penalty system SARSA- λ (State-action-reward-state-action).
- Initial Temporal Difference (TD) showed high fluctuation, showing the agent's early exploration of suboptimal routes.
- Over time, the learning curve smoothed out, depicting convergence to a stable and effective routing policy.

Anomaly Detection with Isolation Forest (Fig. 8):

- Detects outliers in communication delays and unusual spikes/dips in all 3 links.

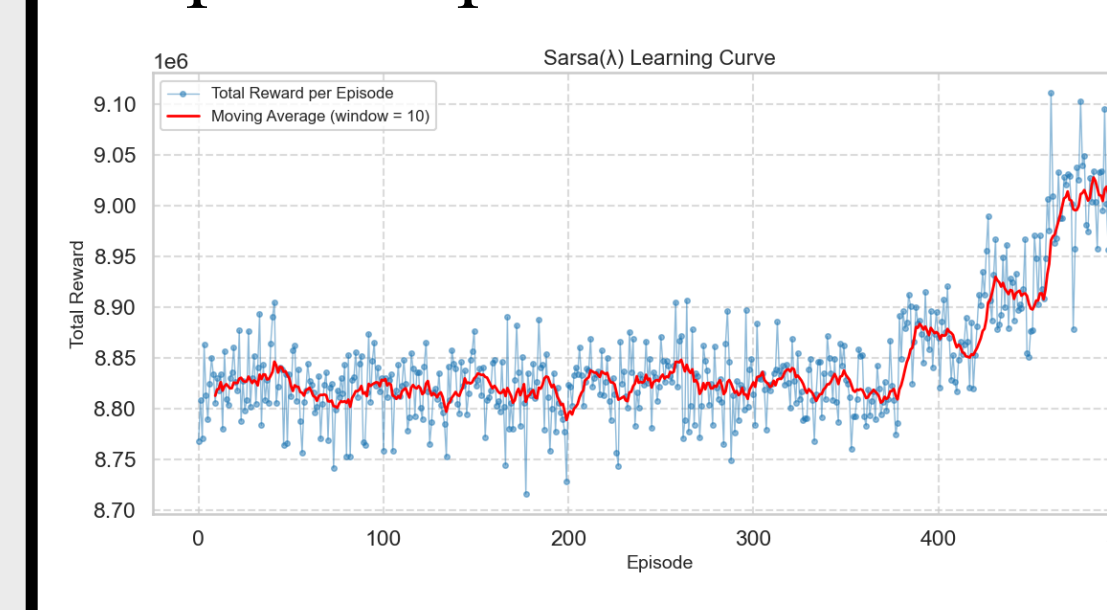


Fig. 7: Learning curve of RL based DTN

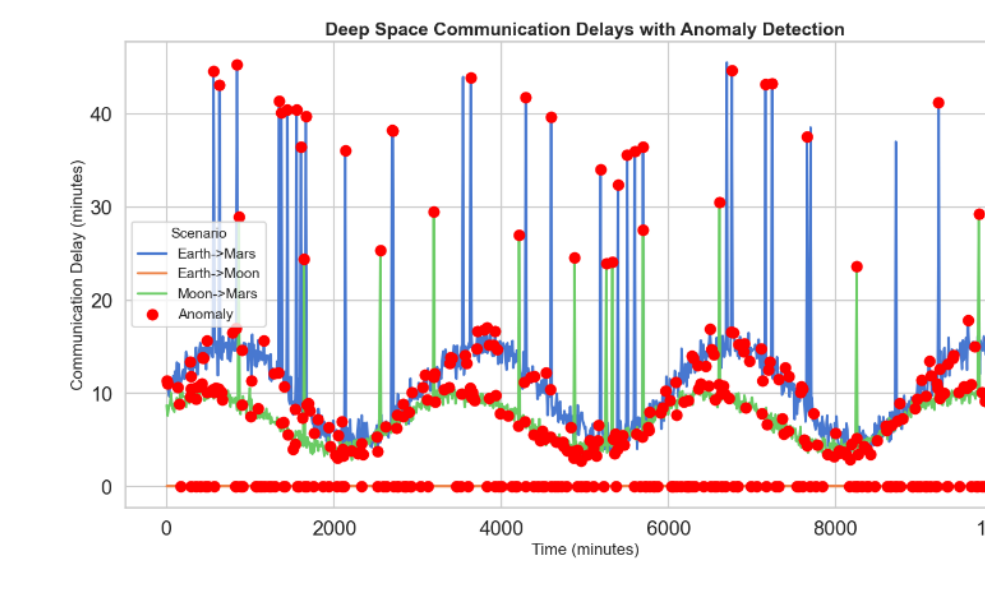


Fig. 8: Anomaly Detection using Isolation Forest

Packet Delivery Ratio (PDR) & Overhead Ratio Analysis (Fig. 9):

- RL-DTN: Balanced performance, better than Epidemic & PROPHET in both areas.
- RL-DTN is efficient without rigid scheduling (CGR) or vast labelled data (CDT).

Hamming Loss Comparison (Fig. 10):

- The CGR approach results in a Hamming Loss of 0.20, while the RL-DTN achieves 0.22 despite having no prior schedule awareness.

RL-DTN validates learning-based approaches: Despite not knowing future contact opportunities, RL-DTN closely matches the performance of CGR, demonstrating the potential of reinforcement learning in DTNs.

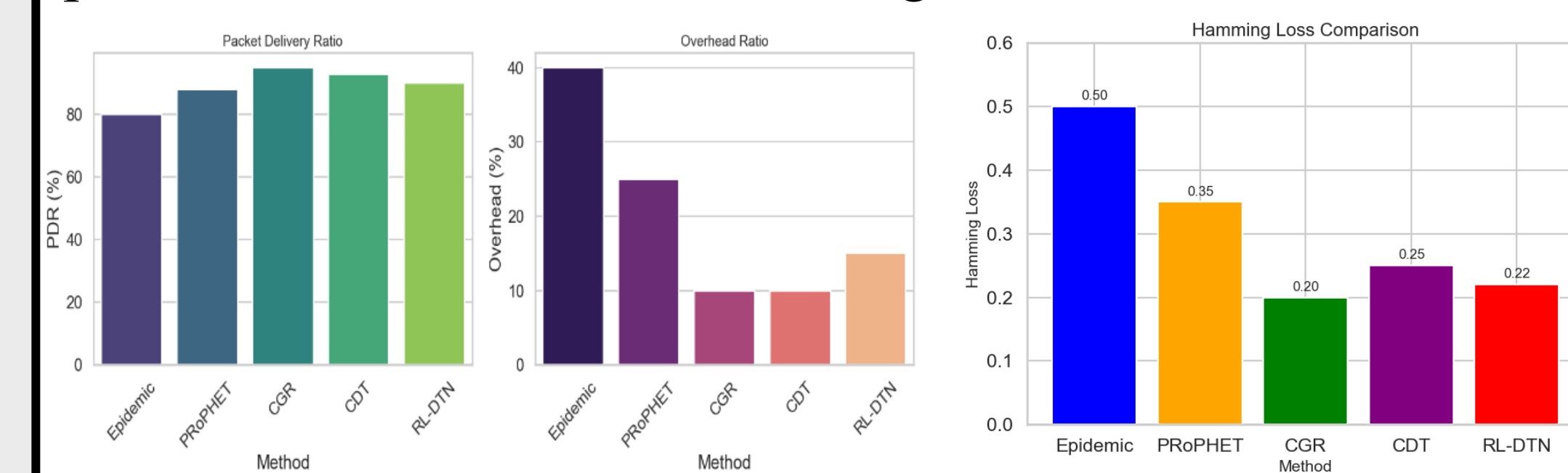


Fig. 9: Comparison of RL-DTN against existing models

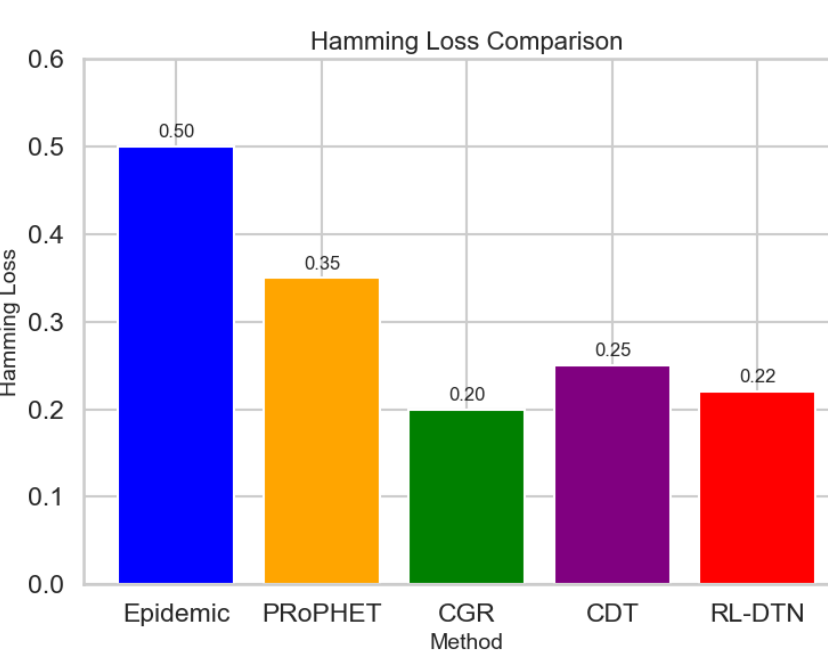


Fig. 10: Comparison of Hamming Loss between existing models

CONCLUSION

This project presents a comprehensive analysis of a Ka-Band and IP Stack based communication system which utilizes reinforcement learning and isolation forest algorithms for optimal routing and anomaly detection. The Monte Carlo simulations carried out on MATLAB prove the high gain antenna to be a suitable system. The RL model is proven to be able to train itself to find the most optimal routes. The isolation forest algorithm is successfully able to isolate and detection anomalies in the synthetic dataset. IPv6 and SCPS TP protocols were successfully demonstrated with SCPS-TP performing better than LTP in deep space conditions and RL-DTN is shown to be a balance between existing DTN protocols.

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CONFERENCE PUBLICATION

Our paper entitled 'Deep Space Communication utilizing IP Stack and Reinforcement Learning Algorithm' was presented at IEEE co-sponsored International Conference on Wireless Communications, Signal Processing and Networking (WISPNET) 2025 on 22nd March 2025.